

ECEN-662: Homework 6

Due on March 25, 2026 at 11:59 PM

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Problem 1

Consider a Gaussian Mixture Model (GMM) with two well-separated modes:

$$p(\theta) = 0.5 \cdot \mathcal{N}(\theta; -3, 1) + 0.5 \cdot \mathcal{N}(\theta; 3, 1),$$

where $\mathcal{N}(\theta; \mu, \sigma^2)$ is a Gaussian density with mean μ and variance σ^2 . Our goal is to approximate the above GMM using the variational distribution:

$$q_\lambda(\theta) = \mathcal{N}(\theta; \mu, \sigma^2),$$

where the variational parameter $\lambda = (\mu, \sigma)$.

- i) Derive the explicit update rule for both μ and σ using the re-parameterization trick.
- ii) Write a computer program to find the optimal variational distribution for the above GMM model. Try multiple initial values for μ and σ and describe how the value of μ and σ change over the iterations.
- iii) Repeat Part ii) for the following updated GMM model:

$$p(\theta) = 0.5 \cdot \mathcal{N}(\theta; -3, 4) + 0.5 \cdot \mathcal{N}(\theta; 3, 0.25).$$

Solution

Part (i): ELBO:

$$\mathcal{L}(\mu, \sigma) = \mathbb{E}_{q_\lambda} [\log q_\lambda(\theta) - \log p(\theta)]$$

So

$$\mathcal{L}(\mu, \sigma) = \mathbb{E}_{\epsilon \sim \mathcal{N}(0,1)} [\log q_\lambda(\mu + \sigma\epsilon) - \log p(\mu + \sigma\epsilon)]$$

Which becomes

$$\mathcal{L}(\mu, \sigma) = -\frac{1}{2} \log(2\pi\sigma^2) - \mathbb{E}_{\epsilon \sim \mathcal{N}(0,1)} [\log p(\mu + \sigma\epsilon)]$$

Now taking the gradients:

$$\begin{aligned} \nabla_\mu \mathcal{L}(\mu, \sigma) &= -\mathbb{E}_{\epsilon \sim \mathcal{N}(0,1)} [\nabla_\theta \log p(\theta)] \\ \nabla_\sigma \mathcal{L}(\mu, \sigma) &= -\frac{1}{\sigma} - \mathbb{E}_{\epsilon \sim \mathcal{N}(0,1)} [\epsilon \cdot \nabla_\theta \log p(\theta)] \end{aligned}$$

Thus we update μ and σ as follows:

$$\begin{aligned} \mu &\leftarrow \mu + \alpha \nabla_\mu \mathcal{L}(\mu, \sigma) \\ \sigma &\leftarrow \sigma + \alpha \left(\frac{1}{\sigma} + \mathbb{E}_{\epsilon \sim \mathcal{N}(0,1)} [\epsilon \cdot \nabla_\theta \log p(\mu + \sigma\epsilon)] \right) \end{aligned}$$

Part (ii):

Initial μ	Initial σ	Final (μ, σ)
-3	1	(-3.04, 1.01)
-3	3	(-3.02, 1.01)
0	1	(-0.13, 2.74)
0	3	(0.16, 2.84)
3	1	(3.06, 1.00)
3	3	(2.96, 1.03)

Part (iii): Rerunning but now with

$$p(\theta) = 0.5 \cdot \mathcal{N}(\theta; -3, 4) + 0.5 \cdot \mathcal{N}(\theta; 3, 0.25)$$

yields

Initial μ	Initial σ	Final (μ, σ)
-3	1	(-2.04, 2.79)
-3	3	(-1.96, 2.62)
0	1	(-2.06, 2.82)
0	3	(-2.09, 2.62)
3	1	(3.01, 0.51)
3	3	(3.01, 0.51)